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2.2.30. SIZE-EXCLUSION CHROMATOGRAPHY

PRINCIPLE

Size-exclusion chromatography is a liquid chromatography (2.2.29) technique that separates molecules in solution according to their size. With organic mobile phases the technique is known as gel-permeation chromatography and with aqueous mobile phases it is known as gel-filtration chromatography. The sample is introduced into a column, which is filled with a gel or a porous particle packing material, and is carried by the mobile phase through the column. The size separation takes place by repeated exchange of the solute molecules between the solvent of the mobile phase and the same solvent in the stagnant liquid phase (stationary phase) within the pores of the packing material. The pore-size range of the packing material determines the molecular-size range within which separation can occur.

Molecules small enough to penetrate all the pore spaces elute at the total mobile phase volume V_t (also known as total permeation volume). Molecules apparently larger than the maximum pore size of the packing material migrate along the column only through the spaces between the particles of the packing material without being retained and elute at the retention volume of an unretained compound V_0 (also known as exclusion volume or void volume). Separation according to molecular size occurs between the retention volume of an unretained compound and the total mobile phase volume, with useful separation usually occurring in the first two thirds of this range.

EQUIPMENT

The equipment consists essentially of a chromatographic column of varying length and internal diameter ($\emptyset$), if necessary temperature-controlled, packed with a separation material that is capable of fractionation in the appropriate range of molecular sizes. The packing material may be a soft support such as a swollen gel or a rigid support composed of a material such as glass, silica or a solvent-compatible, cross-linked organic polymer. Rigid supports usually require pressurised systems giving faster separations.

One end of the column is usually fitted with a suitable device for applying the sample, such as a flow adapter, a syringe through a septum or an injection valve, and may also be connected to a suitable pump for controlling the flow of the eluent. Alternatively, the sample may be applied directly to the drained bed surface or, where the sample is denser than the eluent, it may be layered beneath the eluent.

The mobile phase is chosen according to sample type, separation medium and technique of detection. The eluent is passed through the column at a constant rate.

The outlet of the column is usually connected to a suitable detector fitted with an automatic recorder that enables the monitoring of the relative concentrations of separated components of the sample. Ultraviolet and visible (UV-Vis) spectrophotometers (2.2.25), differential refractometers (RI), luminescent detectors, multi-angle light-scattering (MALS) detectors and other detectors may be used. An automatic fraction collector may be attached, if necessary.

PROCEDURE

Before carrying out the separation, the packing material is treated and the column packed as described in the monograph, or according to the manufacturer's instructions.

Criteria for assessing the suitability of the system are described in general chapter 2.2.46. *Chromatographic separation techniques*. The extent to which adjustments of chromatographic conditions can be made are also given in that general chapter.

DETERMINATION OF RELATIVE COMPONENT COMPOSITION OF MIXTURES

Carry out the separation as stated in the monograph. If possible, monitor the elution of the components continuously and measure the corresponding peak areas. If the sample is monitored by a physico-chemical property to which all the components of interest exhibit equivalent responses (for example if they have the same specific absorbance), calculate the relative amount of each component by dividing the respective peak area by the sum of the peak areas of all the components of interest. If the responses to the property used for detection of the components of interest are not equivalent, calculate the content by means of calibration curves obtained with the calibration standards prescribed in the monograph.

DETERMINATION OF MOLECULAR MASSES

Size-exclusion chromatography may be used to determine molecular masses by comparison with appropriate calibration standards specified in the monograph. The retention volumes of the calibration standards may be plotted against the logarithm of their molecular masses. The plot usually approximates a straight line within the exclusion and total permeation limits for the separation medium used. From the calibration curve, molecular masses may be estimated. The molecular-mass calibration is valid only for the particular macromolecular solute/solvent system used under the specified experimental conditions.

DETERMINATION OF MOLECULAR SIZE DISTRIBUTION OF POLYMERS

Size-exclusion chromatography may be used to determine the distribution of the molecular size of polymers. However, sample comparison may be valid only for results obtained under the same experimental conditions. The reference standards used for the calibration and the methods for determination of the distribution of molecular sizes of polymers are specified in the monograph.

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2.2.45. SUPERCRITICAL FLUID CHROMATOGRAPHY

PRINCIPLE

Supercritical fluid chromatography (SFC) is a chromatographic separation technique in which the mobile phase is a fluid in a supercritical or a subcritical state. The stationary phase, contained in a column, consists of either finely divided solid particles, such as a silica or porous graphite, a chemically modified stationary phase, as used in liquid chromatography, or, for capillary columns, a cross-linked liquid film evenly coated on the walls of the column.

SFC is based on mechanisms of adsorption or mass distribution.

EQUIPMENT

The apparatus usually consists of a cooled pumping system, an injector, a chromatographic column, contained in an oven, a detector, a pressure regulator and a data acquisition device (or an integrator or a chart recorder).

Pumping system

Pumping systems are required to deliver the mobile phase at a constant flow rate. Pressure fluctuations are to be minimised, e.g. by passing the pressurised solvent through a